

PAMBAN, India

WMO: 433630

Lat: 9.283N

Lon: 79.217E

Elev: 36

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	75.3	76.4	66.8	98.8	78.9	68.1	103.4	79.4	27.0	80.4	24.1	79.0	3.5	N/A	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.2	94.5	82.9	93.5	82.7	92.7	82.4	86.1	90.7	85.1	90.2	84.4	89.7	4.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.9	184.0	89.2	83.8	177.3	88.6	83.0	172.6	88.2	50.8	90.6	49.6	90.3	48.6	89.8	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	20.4	18.0	71.8	98.2	2.7	2.2	69.9	99.7	68.4	101.0	66.9	102.3	65.0	103.8	
			DB												
			WB	67.6	87.0	2.4	2.1	65.9	88.5	64.5	89.7	63.2	90.9	61.5	92.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.1	80.5	82.2	85.4	88.6	88.4	87.9	87.5	86.5	86.0	84.7	82.4	80.7
	DBStd	3.28	1.36	1.72	1.83	1.65	1.84	1.32	1.30	1.37	1.27	2.37	2.07	1.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12801	947	901	1096	1157	1191	1138	1162	1130	1080	1075	972	952
	CDD65	7326	482	481	631	707	726	688	697	665	630	610	522	487
	CDH74	91750	4476	5056	7974	9978	10217	9598	9487	8718	8062	7543	5814	4827
	CDH80	41148	670	1368	3590	5667	5763	5278	5024	4263	3752	3176	1717	880
Wind	WSAvg	6.2	7.7	5.1	3.2	3.7	7.1	8.4	6.5	6.2	6.4	4.9	6.6	8.7
Precipitation	PrecAvg	36.2	1.5	0.9	0.9	2.0	1.9	0.1	0.4	0.4	1.3	7.9	11.0	7.6
	PrecMax	58.9	8.1	12.1	11.4	7.6	7.7	0.8	2.6	2.0	7.8	18.0	20.6	21.8
	PrecMin	14.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	1.5	0.5
	PrecStd	10.2	2.0	1.9	1.8	2.0	2.1	0.2	0.7	0.5	1.9	4.5	4.9	5.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.9	90.3	93.8	95.4	95.3	94.3	94.5	93.6	92.2	92.9	90.0	88.2
		MCWB	77.6	79.5	81.1	83.3	83.6	82.3	81.6	81.7	82.0	81.8	80.8	79.0
	2%	DB	85.1	88.4	92.2	94.6	94.0	93.2	92.8	91.8	90.9	90.9	88.1	85.9
		MCWB	76.8	78.6	80.8	83.1	83.5	82.4	81.7	81.4	81.4	81.3	80.3	78.5
	5%	DB	84.0	86.9	91.1	93.6	93.0	92.3	91.8	90.8	90.0	89.8	86.8	84.5
		MCWB	76.4	78.0	80.5	82.9	83.4	82.4	81.7	81.1	81.2	81.0	79.8	78.1
	10%	DB	83.1	85.6	89.9	92.7	91.9	91.3	90.8	89.7	89.1	88.6	85.6	83.6
		MCWB	76.0	77.5	80.2	82.6	83.1	82.3	81.6	80.8	81.0	80.6	79.2	77.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	80.9	84.1	86.3	86.4	86.6	87.9	85.7	84.6	83.9	82.8	81.6
		MCDB	83.8	87.0	90.3	92.3	91.6	90.7	91.0	90.2	89.8	89.5	87.3	84.3
	2%	WB	78.9	80.0	82.8	85.2	85.3	85.4	86.2	83.6	83.1	82.7	81.5	80.1
		MCDB	82.7	86.1	89.2	91.5	90.9	89.9	89.5	88.5	88.8	88.6	86.2	83.6
	5%	WB	78.1	79.2	81.9	84.3	84.5	84.4	84.5	82.5	82.3	81.9	80.7	79.2
		MCDB	82.0	84.9	88.4	90.9	90.3	89.1	87.2	88.1	88.1	87.6	85.3	82.7
	10%	WB	77.3	78.6	81.1	83.6	83.9	83.5	82.9	81.7	81.6	81.3	80.0	78.8
		MCDB	81.5	84.1	87.6	90.2	89.7	88.7	87.6	87.5	87.3	86.8	84.4	82.3
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.9	10.0	9.2	7.8	7.4	7.8	7.8	7.7	7.7	6.2	5.5
		MCDBR	6.9	9.4	11.1	9.6	8.4	8.2	8.6	8.7	8.2	9.0	8.3	7.1
	5% WB	MCWBR	3.6	3.9	4.4	3.5	3.4	3.8	4.0	3.8	3.4	3.6	4.0	3.8
		MCWBR	6.1	8.5	9.9	9.0	7.7	7.7	8.3	8.1	8.1	8.4	7.4	6.3
Clear-Sky Solar Irradiance	taub		0.450	0.447	0.484	0.497	0.563	0.507	0.522	0.506	0.487	0.498	0.493	0.485
		taud	2.308	2.301	2.203	2.209	2.012	2.204	2.146	2.200	2.262	2.223	2.225	2.221
	Ebn at Noon		273	278	268	261	239	250	247	255	264	260	258	259
		Edh at Noon	42	43	48	47	56	46	49	47	44	46	45	44
All-Sky Solar Radiation	RadAvg		1650	1886	2035	1990	1897	1860	1848	1876	1898	1651	1371	1397
	RadStd		100	89	92	87	116	91	70	90	95	101	117	108

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.61	-0.29	N/A	+1.82	N/A	-0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page